MATH 101 **91736**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. (a) Expand and simplify. (b) Factor completely. $(a)\quad (2x-3)(x+2)$ $(b)\quad 2x^2-5x-3$	1. (a) 10 pts (b) 10 pts
2. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-2}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{18}$	2. (a) 10 pts (b) 10 pts
3. Solve both equations. $(a)\quad x^2+2x-3=0$ $(b)\quad \frac{4}{2x-4}+\frac{1}{x-2}=3$	3. (a) 10 pts (b) 10 pts
4. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 3$	4. 20 pts
5. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-4}{x^2+2x-8}$	5. (a) 14 pts (b) 6 pts

91736

TEST PAGE 1

1. (a)

$2x^2 + x - 6$ — 10 pts

(b)

$(x - 3)(2x + 1)$ — 10 pts

2. (a)

$\frac{2y^4}{x^3}$ — 10 pts

(b)

$3\sqrt{2}$ — 10 pts

3. (a)

$x = -3,\ 1$ — 10 pts

(b)

$x = 3$ — 10 pts

4.

$[-1, 2]$ — 20 pts

5. (a)

$\frac{x+2}{x+4}$ — 14 pts

(b)

$\{-4, 2\}$ — 6 pts

MATH 101 **94928**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-4}{x^2+2x-8}$	1. (a) 14 pts (b) 6 pts
2. (a) Expand and simplify. (b) Factor completely. $(a)\quad (2x-3)(x+2)$ $(b)\quad 2x^2-5x-3$	2. (a) 10 pts (b) 10 pts
3. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-2}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{27}$	3. (a) 10 pts (b) 10 pts
4. Solve both equations. $(a)\quad x^2+2x-3=0$ $(b)\quad \frac{4}{2x-4}+\frac{1}{x-2}=3$	4. (a) 10 pts (b) 10 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 3$	5. 20 pts

94928

TEST PAGE 1

1. (a)

$\frac{x+2}{x+4}$ — 14 pts

(b)

$\{-4, 2\}$ — 6 pts

2. (a)

$2x^2 + x - 6$ — 10 pts

(b)

$(x-3)(2x+1)$ — 10 pts

3. (a)

$\frac{2y^4}{x^3}$ — 10 pts

(b)

$3\sqrt{3}$ — 10 pts

4. (a)

$x = -3,\ 1$ — 10 pts

(b)

$x = 3$ — 10 pts

5.

$[-1, 2]$

20 pts

MATH 101 **37956**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. (a) Expand and simplify. (b) Factor completely. $(a)\quad (2x-3)(x+2)$ $(b)\quad 2x^2-5x-3$	1. (a) 10 pts (b) 10 pts
2. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 3$	2. 20 pts
3. Solve both equations. $(a)\quad x^2+2x-3=0$ $(b)\quad \frac{4}{2x-4}+\frac{1}{x-2}=3$	3. (a) 10 pts (b) 10 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-2}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{45}$	4. (a) 10 pts (b) 10 pts
5. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-4}{x^2+2x-8}$	5. (a) 14 pts (b) 6 pts

37956

TEST PAGE 1

1. (a)

$2x^2 + x - 6$ 10 pts

(b)

$(x - 3)(2x + 1)$ 10 pts

2.

$[-1, 2]$

20 pts

3. (a)

$x = -3,\ 1$ 10 pts

(b)

$x = 3$ 10 pts

4. (a)

$\frac{2y^4}{x^3}$ 10 pts

(b)

$3\sqrt{5}$ 10 pts

5. (a)

$\frac{x+2}{x+4}$ 14 pts

(b)

$\{-4, 2\}$ 6 pts

MATH 101 **26374**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify. In (a), x and y are nonzero; give positive exponents. (a) $\frac{6x^{-3}y^{3}}{3xy^{-1}}$ (b) $\sqrt{32}$	1. (a) 10 pts (b) 10 pts
2. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 4$	2. 20 pts
3. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^{2}-9}{x^{2}+2x-15}$	3. (a) 14 pts (b) 6 pts
4. (a) Expand and simplify. (b) Factor completely. (a) $(3x-3)(x+3)$ (b) $2x^{2}-6x-8$	4. (a) 10 pts (b) 10 pts
5. Solve both equations. (a) $x^{2}+2x-8=0$ (b) $\frac{5}{2x-6}+\frac{1}{x-3}=\frac{7}{2}$	5. (a) 10 pts (b) 10 pts

26374

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^4}$ 10 pts

(b)

$4\sqrt{2}$ 10 pts

2.

$[-\frac{3}{2}, \frac{5}{2}]$ 20 pts

3. (a)

$\frac{x+3}{x+5}$ 14 pts

(b)

$\{-5, 3\}$ 6 pts

4. (a)

$3x^2 + 6x - 9$ 10 pts

(b)

$2(x-4)(x+1)$ 10 pts

5. (a)

$x = -4,\ 2$ 10 pts

(b)

$x = 4$ 10 pts

MATH 101 **60439**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. (a) Expand and simplify. (b) Factor completely. $(a)\quad (3x-3)(x+3)$ $(b)\quad 2x^2-6x-8$	1. (a) 10 pts (b) 10 pts
2. Solve both equations. $(a)\quad x^2+2x-8=0$ $(b)\quad \frac{5}{2x-6}+\frac{1}{x-3}=\frac{7}{2}$	2. (a) 10 pts (b) 10 pts
3. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-9}{x^2+2x-15}$	3. (a) 14 pts (b) 6 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-3}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{48}$	4. (a) 10 pts (b) 10 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 4$	5. 20 pts

60439

TEST PAGE 1

1. (a)

$3x^2 + 6x - 9$ — 10 pts

(b)

$2(x-4)(x+1)$ — 10 pts

2. (a)

$x = -4,\ 2$ — 10 pts

(b)

$x = 4$ — 10 pts

3. (a)

$\frac{x+3}{x+5}$ — 14 pts

(b)

$\{-5, 3\}$ — 6 pts

4. (a)

$\frac{2y^4}{x^4}$ — 10 pts

(b)

$4\sqrt{3}$ — 10 pts

5.

$[-\frac{3}{2}, \frac{5}{2}]$ — 20 pts

MATH 101 **62223**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve both equations. $(a)\quad x^2+2x-8=0$ $(b)\quad \frac{5}{2x-6}+\frac{1}{x-3}=\frac{7}{2}$	1. (a) 10 pts (b) 10 pts
2. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-9}{x^2+2x-15}$	2. (a) 14 pts (b) 6 pts
3. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 4$	3. 20 pts
4. (a) Expand and simplify. (b) Factor completely. $(a)\quad (3x-3)(x+3)$ $(b)\quad 2x^2-6x-8$	4. (a) 10 pts (b) 10 pts
5. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-3}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{80}$	5. (a) 10 pts (b) 10 pts

62223

TEST PAGE 1

1. (a)

$x = -4,\ 2$ — 10 pts

(b)

$x = 4$ — 10 pts

2. (a)

$\frac{x+3}{x+5}$ — 14 pts

(b)

$\{-5, 3\}$ — 6 pts

3.

$[-\frac{3}{2}, \frac{5}{2}]$ — 20 pts

4. (a)

$3x^2 + 6x - 9$ — 10 pts

(b)

$2(x-4)(x+1)$ — 10 pts

5. (a)

$\frac{2y^4}{x^4}$ — 10 pts

(b)

$4\sqrt{5}$ — 10 pts

MATH 101 **85650**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the inequality and write the solution in interval notation. $\|2x-1\| \le 5$	1. 20 pts
2. Solve both equations. $(a)\quad x^2+2x-15=0$ $(b)\quad \frac{6}{2x-8}+\frac{1}{x-4}=4$	2. (a) 10 pts (b) 10 pts
3. (a) Expand and simplify. (b) Factor completely. $(a)\quad (4x-3)(x+4)$ $(b)\quad 2x^2-7x-15$	3. (a) 10 pts (b) 10 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-4}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{50}$	4. (a) 10 pts (b) 10 pts
5. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-16}{x^2+2x-24}$	5. (a) 14 pts (b) 6 pts

85650

TEST PAGE 1

1.

$[-2, 3]$

20 pts

2. (a)

$x = -5,\ 3$ 10 pts

(b)

$x = 5$ 10 pts

3. (a)

$4x^2 + 13x - 12$ 10 pts

(b)

$(x - 5)(2x + 3)$ 10 pts

4. (a)

$\frac{2y^4}{x^5}$ 10 pts

(b)

$5\sqrt{2}$ 10 pts

5. (a)

$\frac{x+4}{x+6}$ 14 pts

(b)

$\{-6, 4\}$ 6 pts

MATH 101 **51783**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve both equations. $(a)\quad x^2+2x-15=0$ $(b)\quad \frac{6}{2x-8}+\frac{1}{x-4}=4$	1. (a) 10 pts (b) 10 pts
2. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-16}{x^2+2x-24}$	2. (a) 14 pts (b) 6 pts
3. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 5$	3. 20 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-4}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{75}$	4. (a) 10 pts (b) 10 pts
5. (a) Expand and simplify. (b) Factor completely. $(a)\quad (4x-3)(x+4)$ $(b)\quad 2x^2-7x-15$	5. (a) 10 pts (b) 10 pts

51783

TEST PAGE 1

1. (a)

$x = -5,\ 3$ 10 pts

(b)

$x = 5$ 10 pts

2. (a)

$\frac{x+4}{x+6}$ 14 pts

(b)

$\{-6, 4\}$ 6 pts

3.

$[-2, 3]$

20 pts

4. (a)

$\frac{2y^4}{x^5}$ 10 pts

(b)

$5\sqrt{3}$ 10 pts

5. (a)

$4x^2 + 13x - 12$ 10 pts

(b)

$(x-5)(2x+3)$ 10 pts

MATH 101 **90916**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify. In (a), x and y are nonzero; give positive exponents. (a) $\frac{6x^{-4}y^{3}}{3xy^{-1}}$ (b) $\sqrt{125}$	1. (a) 10 pts (b) 10 pts
2. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 5$	2. 20 pts
3. Solve both equations. (a) $x^{2}+2x-15=0$ (b) $\frac{6}{2x-8}+\frac{1}{x-4}=4$	3. (a) 10 pts (b) 10 pts
4. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^{2}-16}{x^{2}+2x-24}$	4. (a) 14 pts (b) 6 pts
5. (a) Expand and simplify. (b) Factor completely. (a) $(4x-3)(x+4)$ (b) $2x^{2}-7x-15$	5. (a) 10 pts (b) 10 pts

90916

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^5}$ 10 pts

(b)

$5\sqrt{5}$ 10 pts

2.

$[-2, 3]$

20 pts

3. (a)

$x = -5,\ 3$ 10 pts

(b)

$x = 5$ 10 pts

4. (a)

$\frac{x+4}{x+6}$ 14 pts

(b)

$\{-6, 4\}$ 6 pts

5. (a)

$4x^2 + 13x - 12$ 10 pts

(b)

$(x-5)(2x+3)$ 10 pts

MATH 101 **74581**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-25}{x^2+2x-35}$	1. (a) 14 pts (b) 6 pts
2. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-5}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{72}$	2. (a) 10 pts (b) 10 pts
3. (a) Expand and simplify. (b) Factor completely. $(a)\quad (5x-3)(x+5)$ $(b)\quad 2x^2-8x-24$	3. (a) 10 pts (b) 10 pts
4. Solve both equations. $(a)\quad x^2+2x-24=0$ $(b)\quad \frac{7}{2x-10}+\frac{1}{x-5}=\frac{9}{2}$	4. (a) 10 pts (b) 10 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 6$	5. 20 pts

74581

TEST PAGE 1

1. (a)

$\frac{x+5}{x+7}$ — 14 pts

(b)

$\{-7, 5\}$ — 6 pts

2. (a)

$\frac{2y^4}{x^6}$ — 10 pts

(b)

$6\sqrt{2}$ — 10 pts

3. (a)

$5x^2 + 22x - 15$ — 10 pts

(b)

$2(x-6)(x+2)$ — 10 pts

4. (a)

$x = -6,\ 4$ — 10 pts

(b)

$x = 6$ — 10 pts

5.

$[-\frac{5}{2}, \frac{7}{2}]$ — 20 pts

MATH 101 **37126**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the inequality and write the solution in interval notation. $\|2x-1\| \le 6$	1. 20 pts
2. (a) Expand and simplify. (b) Factor completely. $(a)\quad (5x-3)(x+5)$ $(b)\quad 2x^2-8x-24$	2. (a) 10 pts (b) 10 pts
3. Solve both equations. $(a)\quad x^2+2x-24=0$ $(b)\quad \frac{7}{2x-10}+\frac{1}{x-5}=\frac{9}{2}$	3. (a) 10 pts (b) 10 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-5}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{108}$	4. (a) 10 pts (b) 10 pts
5. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-25}{x^2+2x-35}$	5. (a) 14 pts (b) 6 pts

37126

TEST PAGE 1

1.

$[-\frac{5}{2}, \frac{7}{2}]$ 20 pts

2. (a)

$5x^2 + 22x - 15$ 10 pts

(b)

$2(x-6)(x+2)$ 10 pts

3. (a)

$x = -6,\ 4$ 10 pts

(b)

$x = 6$ 10 pts

4. (a)

$\frac{2y^4}{x^6}$ 10 pts

(b)

$6\sqrt{3}$ 10 pts

5. (a)

$\frac{x+5}{x+7}$ 14 pts

(b)

$\{-7, 5\}$ 6 pts

MATH 101 **62465**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the inequality and write the solution in interval notation. $\lvert 2x-1 \rvert \le 6$	1. 20 pts
2. Solve both equations. $(a)\quad x^2+2x-24=0$ $(b)\quad \frac{7}{2x-10}+\frac{1}{x-5}=\frac{9}{2}$	2. (a) 10 pts (b) 10 pts
3. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-5}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{180}$	3. (a) 10 pts (b) 10 pts
4. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-25}{x^2+2x-35}$	4. (a) 14 pts (b) 6 pts
5. (a) Expand and simplify. (b) Factor completely. $(a)\quad (5x-3)(x+5)$ $(b)\quad 2x^2-8x-24$	5. (a) 10 pts (b) 10 pts

62465

TEST PAGE 1

1.

$[-\frac{5}{2}, \frac{7}{2}]$ 20 pts

2. (a)

$x = -6,\ 4$ 10 pts

(b)

$x = 6$ 10 pts

3. (a)

$\frac{2y^4}{x^6}$ 10 pts

(b)

$6\sqrt{5}$ 10 pts

4. (a)

$\frac{x+5}{x+7}$ 14 pts

(b)

$\{-7, 5\}$ 6 pts

5. (a)

$5x^2 + 22x - 15$ 10 pts

(b)

$2(x-6)(x+2)$ 10 pts

MATH 101 **82313**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-36}{x^2+2x-48}$	1. (a) 14 pts (b) 6 pts
2. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-6}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{98}$	2. (a) 10 pts (b) 10 pts
3. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 7$	3. 20 pts
4. (a) Expand and simplify. (b) Factor completely. $(a)\quad (6x-3)(x+6)$ $(b)\quad 2x^2-9x-35$	4. (a) 10 pts (b) 10 pts
5. Solve both equations. $(a)\quad x^2+2x-35=0$ $(b)\quad \frac{8}{2x-12}+\frac{1}{x-6}=5$	5. (a) 10 pts (b) 10 pts

82313

TEST PAGE 1

1. (a)

$\frac{x+6}{x+8}$ 14 pts

(b)

$\{-8, 6\}$ 6 pts

2. (a)

$\frac{2y^4}{x^7}$ 10 pts

(b)

$7\sqrt{2}$ 10 pts

3.

$[-3, 4]$

20 pts

4. (a)

$6x^2 + 33x - 18$ 10 pts

(b)

$(x-7)(2x+5)$ 10 pts

5. (a)

$x = -7,\ 5$ 10 pts

(b)

$x = 7$ 10 pts

MATH 101 **57784**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. (a) Expand and simplify. (b) Factor completely. $(a)\quad (6x-3)(x+6)$ $(b)\quad 2x^2-9x-35$	1. (a) 10 pts (b) 10 pts
2. Solve both equations. $(a)\quad x^2+2x-35=0$ $(b)\quad \frac{8}{2x-12}+\frac{1}{x-6}=5$	2. (a) 10 pts (b) 10 pts
3. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-36}{x^2+2x-48}$	3. (a) 14 pts (b) 6 pts
4. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 7$	4. 20 pts
5. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-6}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{147}$	5. (a) 10 pts (b) 10 pts

57784

TEST PAGE 1

1. (a)

$6x^2 + 33x - 18$ 10 pts

(b)

$(x - 7)(2x + 5)$ 10 pts

2. (a)

$x = -7,\ 5$ 10 pts

(b)

$x = 7$ 10 pts

3. (a)

$\frac{x+6}{x+8}$ 14 pts

(b)

$\{-8, 6\}$ 6 pts

4.

$[-3, 4]$

20 pts

5. (a)

$\frac{2y^4}{x^7}$ 10 pts

(b)

$7\sqrt{3}$ 10 pts

MATH 101 **87759**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. (a) Expand and simplify. (b) Factor completely. $(a)\quad (6x-3)(x+6)$ $(b)\quad 2x^2-9x-35$	1. (a) 10 pts (b) 10 pts
2. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\dfrac{x^2-36}{x^2+2x-48}$	2. (a) 14 pts (b) 6 pts
3. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 7$	3. 20 pts
4. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \dfrac{6x^{-6}y^{3}}{3xy^{-1}}$ $(b)\quad \sqrt{245}$	4. (a) 10 pts (b) 10 pts
5. Solve both equations. $(a)\quad x^2+2x-35=0$ $(b)\quad \dfrac{8}{2x-12}+\dfrac{1}{x-6}=5$	5. (a) 10 pts (b) 10 pts

87759

TEST PAGE 1

1. (a)

$6x^2 + 33x - 18$ 10 pts

(b)

$(x - 7)(2x + 5)$ 10 pts

2. (a)

$\frac{x+6}{x+8}$ 14 pts

(b)

$\{-8, 6\}$ 6 pts

3.

$[-3, 4]$

20 pts

4. (a)

$\frac{2y^4}{x^7}$ 10 pts

(b)

$7\sqrt{5}$ 10 pts

5. (a)

$x = -7,\ 5$ 10 pts

(b)

$x = 7$ 10 pts

MATH 101 **26020**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-7}y^{3}}{3xy^{-1}}$ $(b)\quad \sqrt{128}$	1. (a) 10 pts (b) 10 pts
2. (a) Expand and simplify. (b) Factor completely. $(a)\quad (7x-3)(x+7)$ $(b)\quad 2x^{2}-10x-48$	2. (a) 10 pts (b) 10 pts
3. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^{2}-49}{x^{2}+2x-63}$	3. (a) 14 pts (b) 6 pts
4. Solve both equations. $(a)\quad x^{2}+2x-48=0$ $(b)\quad \frac{9}{2x-14}+\frac{1}{x-7}=\frac{11}{2}$	4. (a) 10 pts (b) 10 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 8$	5. 20 pts

26020

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^8}$ 10 pts

(b)

$8\sqrt{2}$ 10 pts

2. (a)

$7x^2 + 46x - 21$ 10 pts

(b)

$2(x-8)(x+3)$ 10 pts

3. (a)

$\frac{x+7}{x+9}$ 14 pts

(b)

$\{-9, 7\}$ 6 pts

4. (a)

$x = -8,\ 6$ 10 pts

(b)

$x = 8$ 10 pts

5.

$[-\frac{7}{2}, \frac{9}{2}]$ 20 pts

MATH 101 **57444**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-7}y^{3}}{3xy^{-1}}$ $(b)\quad \sqrt{192}$	1. (a) 10 pts (b) 10 pts
2. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 8$	2. 20 pts
3. (a) Expand and simplify. (b) Factor completely. $(a)\quad (7x-3)(x+7)$ $(b)\quad 2x^{2}-10x-48$	3. (a) 10 pts (b) 10 pts
4. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^{2}-49}{x^{2}+2x-63}$	4. (a) 14 pts (b) 6 pts
5. Solve both equations. $(a)\quad x^{2}+2x-48=0$ $(b)\quad \frac{9}{2x-14}+\frac{1}{x-7}=\frac{11}{2}$	5. (a) 10 pts (b) 10 pts

57444

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^8}$ 10 pts

(b)

$8\sqrt{3}$ 10 pts

2.

$[-\frac{7}{2}, \frac{9}{2}]$ 20 pts

3. (a)

$7x^2 + 46x - 21$ 10 pts

(b)

$2(x-8)(x+3)$ 10 pts

4. (a)

$\frac{x+7}{x+9}$ 14 pts

(b)

$\{-9, 7\}$ 6 pts

5. (a)

$x = -8,\ 6$ 10 pts

(b)

$x = 8$ 10 pts

MATH 101 **79377**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-7}y^{3}}{3xy^{-1}}$ $(b)\quad \sqrt{320}$	1. (a) 10 pts (b) 10 pts
2. (a) Expand and simplify. (b) Factor completely. $(a)\quad (7x-3)(x+7)$ $(b)\quad 2x^{2}-10x-48$	2. (a) 10 pts (b) 10 pts
3. Solve both equations. $(a)\quad x^{2}+2x-48=0$ $(b)\quad \frac{9}{2x-14}+\frac{1}{x-7}=\frac{11}{2}$	3. (a) 10 pts (b) 10 pts
4. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^{2}-49}{x^{2}+2x-63}$	4. (a) 14 pts (b) 6 pts
5. Solve the inequality and write the solution in interval notation. $\lvert 2x-1\rvert \le 8$	5. 20 pts

79377

TEST PAGE 1

1. (a)

$\frac{2y^4}{x^8}$ 10 pts

(b)

$8\sqrt{5}$ 10 pts

2. (a)

$7x^2 + 46x - 21$ 10 pts

(b)

$2(x-8)(x+3)$ 10 pts

3. (a)

$x = -8,\ 6$ 10 pts

(b)

$x = 8$ 10 pts

4. (a)

$\frac{x+7}{x+9}$ 14 pts

(b)

$\{-9, 7\}$ 6 pts

5.

$[-\frac{7}{2}, \frac{9}{2}]$ 20 pts

MATH 101 **90683**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the inequality and write the solution in interval notation. $\lvert 2x-1 \rvert \le 9$	1. 20 pts
2. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\dfrac{x^2-64}{x^2+2x-80}$	2. (a) 14 pts (b) 6 pts
3. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \dfrac{6x^{-8}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{162}$	3. (a) 10 pts (b) 10 pts
4. Solve both equations. $(a)\quad x^2+2x-63=0$ $(b)\quad \dfrac{10}{2x-16}+\dfrac{1}{x-8}=6$	4. (a) 10 pts (b) 10 pts
5. (a) Expand and simplify. (b) Factor completely. $(a)\quad (8x-3)(x+8)$ $(b)\quad 2x^2-11x-63$	5. (a) 10 pts (b) 10 pts

90683

TEST PAGE 1

1.

$[-4, 5]$

20 pts

2. (a)

$\frac{x+8}{x+10}$

14 pts

(b)

$\{-10, 8\}$

6 pts

3. (a)

$\frac{2y^4}{x^9}$

10 pts

(b)

$9\sqrt{2}$

10 pts

4. (a)

$x = -9,\ 7$

10 pts

(b)

$x = 9$

10 pts

5. (a)

$8x^2 + 61x - 24$

10 pts

(b)

$(x - 9)(2x + 7)$

10 pts

MATH 101 **73908**

Unit 1: Expressions, Equations, and Inequalities

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Write final answers in the space provided in the right column.

1. Solve the inequality and write the solution in interval notation. $\|2x-1\| \le 9$	1. 20 pts
2. Solve both equations. $(a)\quad x^2+2x-63=0$ $(b)\quad \frac{10}{2x-16}+\frac{1}{x-8}=6$	2. (a) 10 pts (b) 10 pts
3. Simplify. In (a), x and y are nonzero; give positive exponents. $(a)\quad \frac{6x^{-8}y^3}{3xy^{-1}}$ $(b)\quad \sqrt{243}$	3. (a) 10 pts (b) 10 pts
4. (a) Expand and simplify. (b) Factor completely. $(a)\quad (8x-3)(x+8)$ $(b)\quad 2x^2-11x-63$	4. (a) 10 pts (b) 10 pts
5. Simplify and state all values excluded by the original expression. Record: (a) expression; (b) excluded values. $\frac{x^2-64}{x^2+2x-80}$	5. (a) 14 pts (b) 6 pts

73908

TEST PAGE 1

1.

$[-4, 5]$

20 pts

2. (a)

$x = -9,\ 7$ 10 pts

(b)

$x = 9$ 10 pts

3. (a)

$\frac{2y^4}{x^9}$ 10 pts

(b)

$9\sqrt{3}$ 10 pts

4. (a)

$8x^2 + 61x - 24$ 10 pts

(b)

$(x - 9)(2x + 7)$ 10 pts

5. (a)

$\frac{x + 8}{x + 10}$ 14 pts

(b)

$\{-10, 8\}$ 6 pts